

Step 1: Compute $\Delta_{\min}$

Verified

$$\Delta_{\min} = z \cdot \sqrt{(2p(1-p))/N} \quad (\S 3.4)$$

Step 2: Estimate checkpoint range

Verified

Run a few checkpoints, record min-to-max spread

Range < $\Delta_{\min}$?

Yes

No

At your evaluation budget N

Step 3: Use noise-robust aggregation

Tested §5.2

e.g. uniform averaging, model soup

Argmax may be viable

But verify statistical power first

♦ tested in §5.2: gap below $\Delta_{\min}$
(argmax 50.0% vs soup 48.5%)

Derived from §3 (Steps 1-2) and §5.2 (Step 3) of this paper